25900311 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCCMA200 - AI TECHNIQUES FOR DATA ANALYSIS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. What is semi-structured data? Write an example. [K] / [CO1]
2. Define Data Analysis. [K] / [CO1]
3. State the difference between 1D and 2D array in Numpy. [K] / [CO2]
4. What are the reserved keywords? Give examples. [K] / [CO2]
5. Name the process of converting long-format data into wide-format data in Pandas. Give an example. [U] / [CO3]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **5** marks

6. List the secondary sources of data, describe how each can be collected, and discuss their potential limitations. [U] / [CO1]

7. Describe some environmental impact of AI and how can it be reduced. [U] / [CO1]

8. Explain the built-in list functions with examples. [U] / [CO2]

9. Explain any three data analysis functions in Pandas with examples. [U] / [CO2]

10. What is data interpretation in the data analysis life cycle? Why is the correct interpretation of analysis results important for organizations? [U] / [CO3]

11. Why is data summarization important in data analysis, and how can it be performed? [U] / [CO3]

12. What is missing data? List any two reasons why missing data may occur. Give one example. [K] / [CO3]

[5x5 = 25]

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

13. Explain if, if-else, nested if, if –elif ladder statements in Python with examples. [U] / [CO2]

14. Illustrate with example how to create a grouped bar plot in Matplotlib with a suitable data set and interpret the plot. [U] / [CO3]

[15x1 = 15]